

- **Function**

Introduction of function

A function is a relation between a set of inputs and a set of possible outputs where each input is related to exactly one output.

Definition:

A function f from set A to set B is a rule that assigns to each element x in A exactly one element $f(x)$ in B .
Notation:

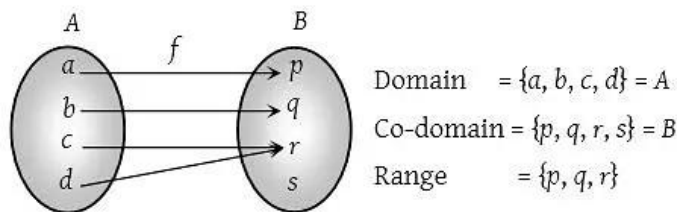
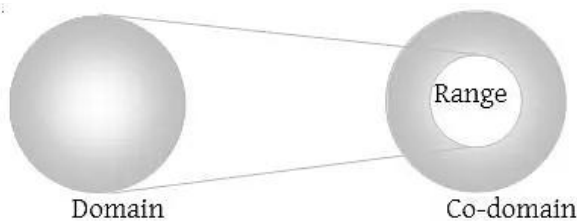
$$f: A \rightarrow B$$

$$f(x) = y$$

Domain: The domain of a function is the set of all valid inputs for which the function is defined. For example, in the function $f(x) = x^2$, the domain is all real numbers (or any number you can plug in for x)

Co domain: The co domain is the set that the function is supposed to map into. It's the set where the function's outputs could potentially lie. For the function $f(x) = x^2$, if you define the co domain as all non-negative real numbers, then the function could produce any value within that set.

Range (or Image): The range (or image) is the subset of the co domain that contains only the actual output values produced by the function when applied to its domain. For the function $f(x) = x^2$, if the domain is all real numbers, then the range would be all non-negative real numbers. The range is the "output" side of the function, showing what the function actually produces.



Let N be the set of natural numbers and the relation R be defined on N such that $R = \{(x, y) : y = 2x, x, y \in N\}$. What is the domain, codomain and range of R ? Is this relation a function?

Given that $y = 2x$ & $x, y \in N$

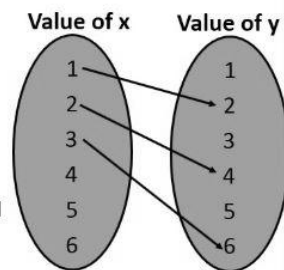
Here, x is always a natural number,

So, **Domain** = Set of natural numbers = N

Here, y is always an even number

Range = Set of even natural numbers

Codomain = Set of Natural numbers = N



• Types of function

Here we shall look at different types of function.

Linear function: A **linear function** has the form of $f(x) = ax + b$ and its graph is a straight line. Since a linear function is a continuous straight line its domain and range would consist of all real numbers. Graphing a linear function can be accomplished by first finding the x- and y-intercepts. This will provide two points that can be used to draw the line.

Finding the intercepts:

1. To find the y-intercept let x equal zero and solve for y
2. To find the x-intercept let y equal zero and solve for x

Example 1: Graph $y = -2x + 6$.

Solution

Step 1: Find the y-intercept

Let $x = 0$ and solve for y

$$y = -2x + 6$$

$$y = -2(0) + 6$$

$$y = 0 + 6$$

$$y = 6$$

The y-intercept is at (0, 6).

Step 2: Find the x-intercept

Let $y = 0$ and solve for x

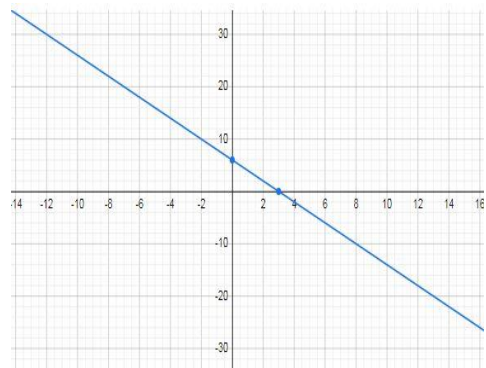
$$y = -2x + 6$$

$$0 = -2x + 6$$

$$-6 = -2x$$

$$3 = x$$

The x-intercept is at (3, 0).



Quadratic function: A **quadratic function** is one of the form $f(x) = ax^2 + bx + c$, where **a**, **b**, and **c** are numbers with **a** not equal to zero. The graph of a quadratic function is a curve called a **parabola**. Parabolas may open upward or downward and vary in "width" or "steepness", but they all have the same basic "U" shape. The picture below shows three graphs, and they are all parabolas.

Therefore, the line of symmetry is the vertical line $x = -\frac{b}{2a}$. We can use the line of symmetry to find the vertex.

Example 1: Graph $y = x^2$

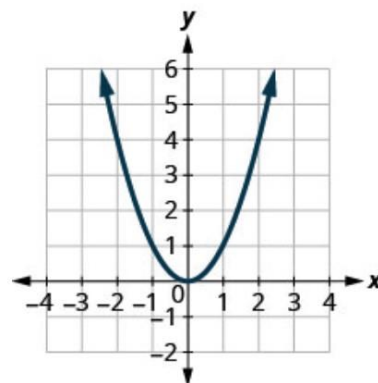
Let $y = 0$ then $x = 0$

$y = 1$ then $x = 1$

$y = 2$ then $x = 4$

$y = -1$ then $x = 1$

$y = -2$ then $x = 4$



Example 2: Graph $y = -x^2 - 2x + 3$

Step 1: Find the y-intercept

Let $x = 0$ and solve for y

$$y = 0 - 0 + 3$$

$$y = 3$$

The y-intercept is at $(0, 3)$.

Step 2: Find the x-intercept

Let $y = 0$ and solve for x

$$0 = -x^2 - 2x + 3$$

$$0 = x^2 + 2x - 3$$

$$0 = (x - 1)(x + 3)$$

$$x = 1 \text{ and } x = -3$$

The x-intercept is at $(1, 0)$ and $(-3, 0)$

Step 3: Find the $x = -\frac{b}{2a}$

$$\text{Let } x = -\frac{b}{2a}, b = -2, a = -1$$

$$x = -\frac{(-2)}{2(-1)}$$

$$x = -1$$

Step 4: Find the y-intercept

Let $x = -1$ and solve for y

$$y = -(-1)^2 - 2(-1) + 3$$

$$y = -1 + 2 + 3$$

$$y = 4$$

The y-intercept is at $(-1, 4)$.

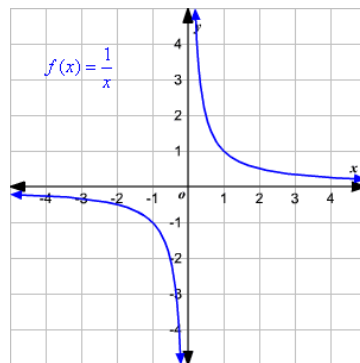
Rational Function: A function is called a rational function if and only if it can be written in the form $f(x) = \frac{P(x)}{Q(x)}$

where P and Q are polynomial functions of x and Q is not the zero function. The domain of f is the set of all values of x for which the denominator $Q(x)$ is not zero.

Example 1: Graph $y = \frac{1}{x}$

x	-4	-3	-2	-1	1	2	3	4
$y = \frac{1}{x}$	$-\frac{1}{4}$	$-\frac{1}{3}$	$-\frac{1}{2}$	-1	1	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$

The parent function of a rational function is $f(x) = \frac{1}{x}$ and the graph is hyperbola.



- (iii) **Polynomial function** A function $f: \mathbf{R} \rightarrow \mathbf{R}$ is said to be *polynomial function* if for each x in $\mathbf{R}$, $y = f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$, where n is a non-negative integer and $a_0, a_1, a_2, \dots, a_n \in \mathbf{R}$.

The functions defined by $f(x) = x^3 - x^2 + 2$, and $g(x) = x^4 + \sqrt{2}x$ are some examples

of polynomial functions, whereas the function h defined by $h(x) = x^{\frac{2}{3}} + 2x$ is not a polynomial function. (Why?)

Example 13 Define the function $f: \mathbf{R} \rightarrow \mathbf{R}$ by $y = f(x) = x^2$, $x \in \mathbf{R}$. Complete the Table given below by using this definition. What is the domain and range of this function? Draw the graph of f .

x	-4	-3	-2	-1	0	1	2	3	4
$y = f(x) = x^2$									

Solution The completed Table is given below:

x	-4	-3	-2	-1	0	1	2	3	4
$y = f(x) = x^2$	16	9	4	1	0	1	4	9	16

Domain of $f = \{x : x \in \mathbf{R}\}$. Range of $f = \{x^2 : x \in \mathbf{R}\}$. The graph of f is given by Fig 2.10

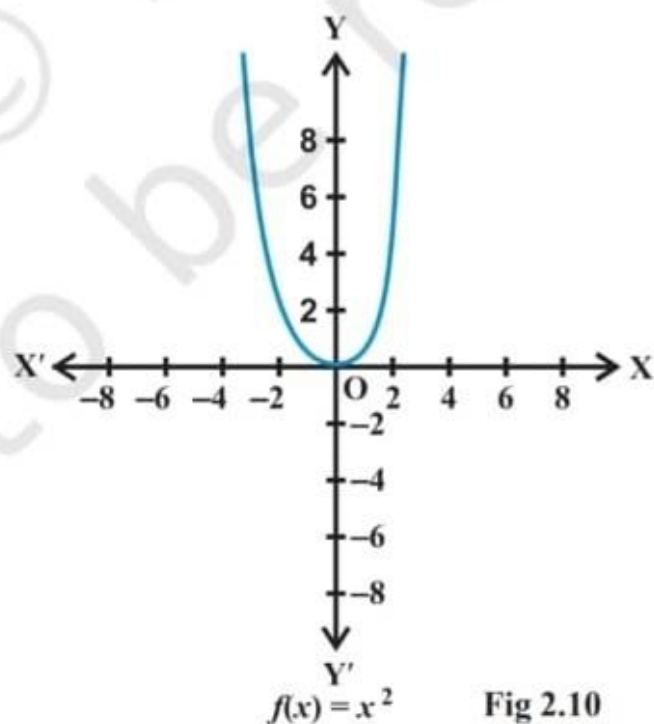


Fig 2.10

(i) **Identity function** Let $\mathbf{R}$ be the set of real numbers. Define the real valued function $f: \mathbf{R} \rightarrow \mathbf{R}$ by $y = f(x) = x$ for each $x \in \mathbf{R}$. Such a function is called the *identity function*. Here the domain and range of f are $\mathbf{R}$. The graph is a straight line as shown in Fig 2.8. It passes through the origin.

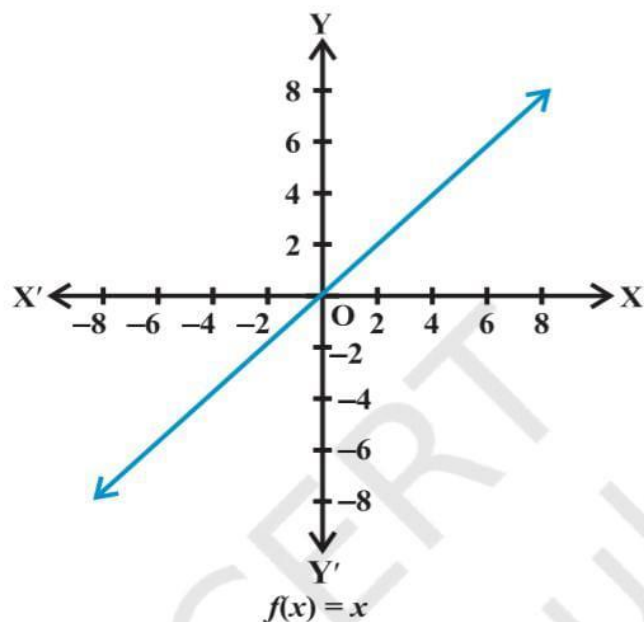
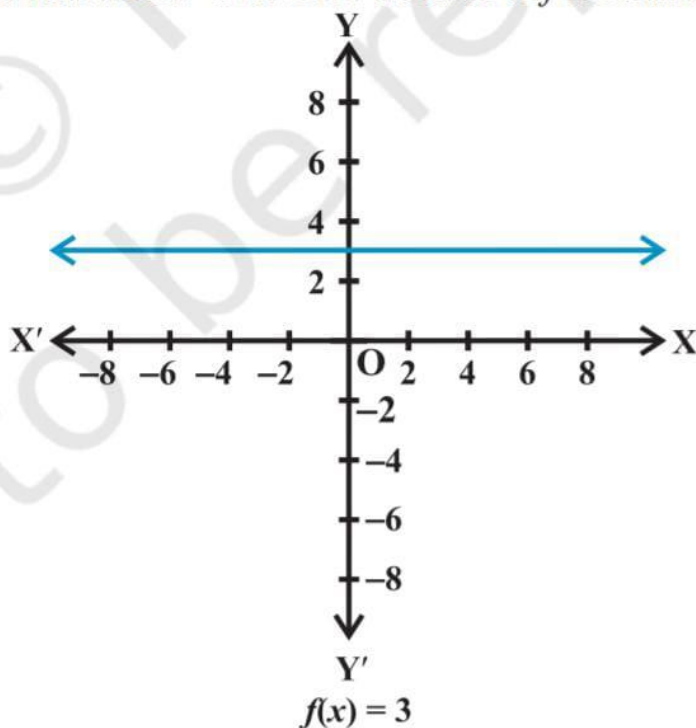


Fig 2.8

(ii) **Constant function** Define the function $f: \mathbf{R} \rightarrow \mathbf{R}$ by $y = f(x) = c$, $x \in \mathbf{R}$ where c is a constant and each $x \in \mathbf{R}$. Here domain of f is $\mathbf{R}$ and its range is $\{c\}$.

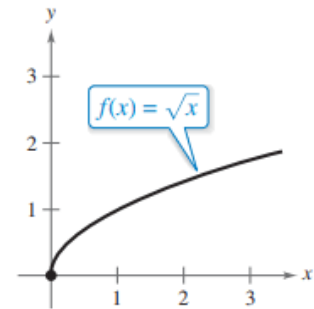


Irrational Function: An Irrational function is one in which the variable x is located under the root function.

Example 1: Graph $f(x) = y = \sqrt{x}$

x	1	2	3	4	5	6	7
y	1	1.414	1.732	2	2.236	2.449	2.645

Ex. $\sqrt{3x+1}$, $\sqrt{x^2-1}$, $\sqrt[3]{x^3-x+1}$



Single value function: A function $f(x)$ is said to be single value function if for each value of x corresponding to just one value of $f(x)$.

$$y = f(x)$$

Example 1: $f(x) = x^2 + 7x - 4$

$$x = 1 \quad f(1) = 1 + 7 - 4$$

$$f(1) = 4$$

$$x = 2 \quad f(2) = 4 + 14 - 4$$

$$f(2) = 14$$

$f(x)$ is single value function.

Example 2: for complex analysis If $w = f(z) = z^2$

$$z = 1, f(1) = 1$$

$$z = 2, f(2) = 4$$

$$z = 3, f(z) = 9$$

$f(z)$ is single value function.

Many or multi value function: A function $f(x)$ is said to be multivalued function if for each value of x corresponds to two or more than two value of $f(n)$.

Example 1: $f(n) = y = n \pm \sqrt{2n^2 - 1}$

$$f(n) = n + \sqrt{2n^2 - 1} \quad \text{and} \quad f(n) = n - \sqrt{2n^2 - 1}$$

$$f(1) = 1 + \sqrt{2(1)^2 - 1} \quad f(1) = 1 - \sqrt{2(1)^2 - 1}$$

$$f(1) = 2 \quad f(1) = 0$$

Example 2: If $w^2 = z$

$$w = \pm\sqrt{z}$$

$$f(z) = w = \pm\sqrt{z}$$

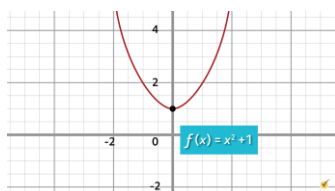
$$f(4) = \pm\sqrt{4} = +2 \text{ and } -2$$

$$f(9) = \pm\sqrt{9} = +3 \text{ and } -3$$

$f(z)$ is many value function.

Even function: A function is 'even' when $f(x) = f(-x)$ for all x .

Ex. $x^2 + 1$ is even function



Odd function: A function is 'even' when $f(x) = -f(x)$ for all x .

Ex. $x^3 + x$

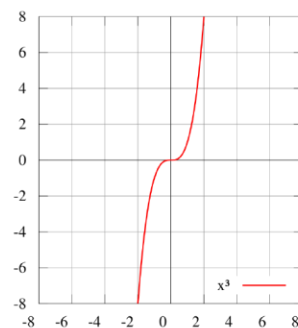
$f(x) = x^3 + x$

$f(-x) = (-x)^3 - x = -x^3 - x = -f(x)$

Ex. $x^5 - 2x^3$

$f(x) = x^5 - 2x^3$

$f(-x) = (-x)^5 - 2(-x)^3 = -x^5 + 2x^3 = -f(x)$



Explicit function: A function in which, dependent variable (say 'y') can be completely expressed in terms of independent variable (say x)

Ex. (i) $y = 5x + 3x - 9$

(ii) $y = \sin x + \cos x$

(iii) $y = \tan^{-1}x + e^x$

In general $y = f(x)$

Implicit function: A function in which, dependent variable say 'y') cannot be completely expressed in terms of independent variable alone.

Ex. (i) $x + xy + y = 0$

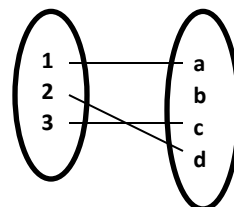
(ii) $\cos(xy) = x + y$ i.e $\cos(xy) - x - y = 0$

(iii) $ye^x + xe^y = 1$ i.e $ye^x + xe^y - 1 = 0$

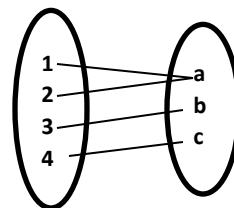
In general $f(x, y) = 0$

• The Classification of function

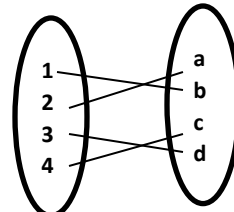
One-one/one to one function: A function is one-one if any two different inputs in the domain correspond to two different outputs in the range. That is if x_1 and x_2 are two different inputs of a function f , then f is one-one if $f(x_1) \neq f(x_2)$



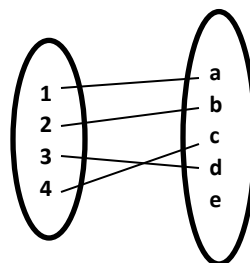
Many one function: Many one function is an important function Which relates two or more elements of the domain set with a single Element of the co-domain set.



Onto function: A function $f:A \rightarrow B$ is called onto if for all b in B there is an a in A such that $f(a) = b$. All elements in co-domain are used.



Into function: If at least one element of B is not the image of an element of A , i.e all the elements of B are not the images, then the function f is called into function.



Example:1 A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = x^2$ is it one one?

$$f(x_1) = f(x_2)$$

$$x_1^2 = x_2^2$$

$$x_1 = x_2 \text{ or } -x_2$$

this function is not one-one.

Example:2 A function $f: \mathbb{R} - \{2\} \rightarrow \mathbb{R} - \{0\}$ is given by $f(x) = \frac{1}{x-2}$ is it one-one?

$$f(x_1) = f(x_2)$$

$$\frac{1}{x_1-2} = \frac{1}{x_2-2}$$

$$x_1 - 2 = x_2 - 2$$

$x_1 = x_2$ is one- one function.

Evaluation of function: To evaluate a function at a given input plug in the input whenever you see a variable in the function then simplify

Ex.1 If $f(x) = 3x^2 + 5x + 1$ find $f(2)$

$$f(x) = 3x^2 + 5x + 1$$

$$f(2x) = 2(2x)^2 + 5(2x) + 1$$

$$f(x) = 23$$

Ex.2 If $f(x) = 2x^2 - 3x + 1$ find $f(2x)$

$$f(x) = 3x^2 + 5x + 1$$

$$f(x) = 3(2)^2 + 5(2) + 1$$

$$f(2x) = 4x^2 - 6x + 1$$

Ex.3 Let $f(x) = \frac{2x-1}{3x+2}$ then simplify $2f(-x)$.

$$2f(-x) = 2 \left[\frac{2(-x)-1}{3(-x)+2} \right]$$

$$2f(-x) = 2 \left[\frac{-2x-1}{-3x+2} \right]$$

$$2f(-x) = \frac{-4x-2}{-3x+2}$$

Ex.4 Evaluate $f(a + b)$ where $f(x) = 4x^2 - 3x$

$$f(a + b) = 4(a + b)^2 - 3(a + b)$$

$$f(a + b) = 4(a^2 + 2ab + b^2) - 3a - 3b$$

$$f(a + b) = 4a^2 + 8ab + 4b^2 - 3a - 3b$$

Ex.5 Evaluate each of the following. If $f(x) = 2x - 1$ and $g(x) = x^2 + x - 2$

Find $(f + g)(2)$, $(f - g)(-1)$, $(fg)(3)$, $\left(\frac{f}{g}\right)(4)$

$$(f + g)(2) = f(2) + g(2)$$

$$(f + g)(2) = 2(2) - 1 + (2)^2 + 2 - 2$$

$$(f + g)(2) = 7$$

$$(f - g)(-1) = f(-1) - g(-1)$$

$$(f - g)(-1) = 2(-1) - 1 - ((-1)^2 + (-1) - 2)$$

$$(f - g)(-1) = -1$$

$$(fg)(3) = (f)(3) \cdot (g)(3)$$

$$(fg)(3) = (2(3) - 1) \cdot ((3)^2 + 3 - 2)$$

$$(fg)(3) = (5) \cdot (10)$$

$$(fg)(3) = 50$$

$$\left(\frac{f}{g}\right)(4) = \frac{f(4)}{g(4)}$$

$$\left(\frac{f}{g}\right)(4) = \frac{2(4)-1}{16+4-2} = \frac{7}{18}$$

Functions - Operations on Functions

Addition $(f + g)(x) = f(x) + g(x)$

Subtraction $(f - g)(x) = f(x) - g(x)$

Multiplication $(f \cdot g)(x) = f(x)g(x)$

Division $(f / g)(x) = f(x) / g(x)$

* Example

Given $f(x) = 2x + 4$, evaluate $f(5)$.

$$f(5) = 2(5) + 4 = 14$$

* Example

$f(x) = x^2 - x - 2$, $g(x) = x + 1$ find $(f + g)(-3)$ OR Evaluate f and g at -3

Evaluate f at -3

$$f(-3) = (-3)^2 - (-3) - 2$$

$$f(-3) = 9 + 3 - 2$$

$$f(-3) = 10$$

Evaluate g at -3

$$g(-3) = (-3) + 1$$

$$g(-3) = -2$$

Now

$$f(-3) + g(-3) = (10) + (-2) = 8$$

* Example

Given $f(x) = x^2$ and $g(x) = 4x + 1$, find $(f + g)(6)$ and $(f - g)(x)$.

$$(f + g)(6) = f(6) + g(6).$$

For the first option, we have $f(6) = 6^2 = 36$ and $g(6) = 4(6) + 1 = 25$.

$$f(6) + g(6) = 36 + 25 = 61.$$

Using the other method,

$$(f + g)(x) = f(x) + g(x) = (x^2) + (4x + 1) = x^2 + 4x + 1.$$

$$(f + g)(6) = f(6) + g(6) = 6^2 + 4(6) + 1 = 61.$$

$$(f - g)(x) = f(x) - g(x).$$

$$(f - g)(x) = f(x) - g(x) = (x^2) - (4x + 1) = x^2 - 4x - 1$$

* Example

Given $f(x) = 4x - 1$, find $2f(x)$.

$$f(x) = 4x - 1$$

$$2f(x) = 2(4x - 1) = 8x - 2$$

* Example

Given $f(x) = 2x$ and $g(x) = x + 1$, find $(fg)(x)$ and $(fg)(3)$.

$$(fg)(x) = f(x)g(x).$$

$$(fg)(x) = f(x)g(x) = (2x) \times (x + 1) = 2x(x + 1) = 2x^2 + 2x$$

$$(fg)(3) = f(3)g(3)$$

$$f(3) = 2 \times 3 = 6, \quad g(3) = 3 + 1 = 4, \quad (fg)(3) = 6 \times 4 = 24$$

$$(fg)(x) = 2x^2 + 2x, \quad (fg)(3) = 2(3)^2 + 2(3) = 18 + 6 = 24$$

* **Example**

Given $f(x) = 3x + 2$ and $g(x) = 4 - 5x$, find $(f + g)(x)$, $(f - g)(x)$, $(f \times g)(x)$, and $(f / g)(x)$

$$\begin{aligned}(f + g)(x) &= f(x) + g(x) \\ &= [3x + 2] + [4 - 5x] \\ &= 3x + 2 + 4 - 5x \\ &= 3x - 5x + 2 + 4 \\ &= -2x + 6\end{aligned}$$

$$\begin{aligned}(f - g)(x) &= f(x) - g(x) \\ &= [3x + 2] - [4 - 5x] \\ &= 3x + 2 - 4 + 5x \\ &= 3x + 5x + 2 - 4 \\ &= 8x - 2\end{aligned}$$

$$\begin{aligned}(f \times g)(x) &= [f(x)][g(x)] \\ &= (3x + 2)(4 - 5x) \\ &= 12x + 8 - 15x^2 - 10x \\ &= -15x^2 + 2x + 8\end{aligned}$$

$$\begin{aligned}\left(\frac{f}{g}\right)(x) &= \frac{f(x)}{g(x)} \\ &= \frac{3x+2}{4-5x}\end{aligned}$$

* **Example**

Given $f(x) = 2x$, $g(x) = x + 4$, and $h(x) = 5 - x^3$, find $(f + g)(2)$, $(h - g)(2)$, $(f \times h)(2)$, and $(h / g)(2)$

$$f(2) = 2(2) = 4$$

$$g(2) = (2) + 4 = 6$$

$$h(2) = 5 - (2)^3 = 5 - 8 = -3$$

Now I can evaluate the listed expressions:

$$\begin{aligned}(f + g)(2) &= f(2) + g(2) \\ &= 4 + 6 = 10\end{aligned}$$

$$\begin{aligned}(h - g)(2) &= h(2) - g(2) \\ &= -3 - 6 = -9\end{aligned}$$

$$\begin{aligned}(f \times h)(2) &= f(2) \times h(2) \\ &= (4)(-3) = -12\end{aligned}$$

$$\begin{aligned}(h / g)(2) &= h(2) \div g(2) \\ &= -3 \div 6 = -0.5\end{aligned}$$

* **Example**

$f(x) = x^2 - 4x - 5$, $g(x) = x - 5$ Find $(f / g)(x)$

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

Replace $f(x)$ with $(x^2 - 4x - 5)$ and $g(x)$ with $(x - 5)$

$$\begin{aligned}\left(\frac{f}{g}\right)(x) &= \frac{x^2 - 4x - 5}{x - 5} \\ &= \frac{(x-5)(x+1)}{(x-5)} \\ &= (x+1)\end{aligned}$$

Composition of function: If f and g are function, then the composite function or composition, f and g is defined by $(f \circ g)(x) = f(g(x))$ or $(g \circ f)(x) = g(f(x))$

Ex.1 $f(x) = 3x + 2$ and $g(x) = x - 3$

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\(f \circ g)(x) &= f(x - 3) \\(f \circ g)(x) &= 3(x - 3) + 2 \\(f \circ g)(x) &= 3x - 9 + 2 \\(f \circ g)(x) &= 3x - 7\end{aligned}$$

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\(g \circ f)(x) &= g(3x + 2) \\(g \circ f)(x) &= 3x + 2 - 3 \\(g \circ f)(x) &= 3x - 1\end{aligned}$$

Ex.2 $f(x) = 2x^2 + 3$ and $g(x) = 3x + 1$

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\(f \circ g)(x) &= f(3x + 1) \\(f \circ g)(x) &= 2(3x + 1)^2 + 3 \\(f \circ g)(x) &= 2(9x^2 + 6x + 1) + 3 \\(f \circ g)(x) &= 18x^2 + 12x + 2 + 3 \\(f \circ g)(x) &= 18x^2 + 12x + 5\end{aligned}$$

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\(g \circ f)(x) &= g(2x^2 + 3) \\(g \circ f)(x) &= 3(2x^2 + 3) + 1 \\(g \circ f)(x) &= 6x^2 + 9 + 1 \\(g \circ f)(x) &= 6x^2 + 10\end{aligned}$$

Ex.3 $f(x) = \frac{1}{x+3}$ and $g(x) = \frac{x}{x-2}$

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\(f \circ g)(x) &= f\left(\frac{x}{x-2}\right) \\(f \circ g)(x) &= \frac{1}{\frac{x}{x-2} + 3} \\(f \circ g)(x) &= \frac{1(x-2)}{x + (x-2)(3)} \\(f \circ g)(x) &= \frac{x-2}{x + 3x - 6} \\(f \circ g)(x) &= \frac{x}{4x - 6} \\(f \circ g)(x) &= \frac{x}{2(2x - 3)}\end{aligned}$$

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\(g \circ f)(x) &= g\left(\frac{1}{x+3}\right) \\(g \circ f)(x) &= \frac{\frac{1}{x+3}}{\frac{1}{x+3} - 2} \\(g \circ f)(x) &= \frac{1}{1 - 2(x+3)} \\(g \circ f)(x) &= \frac{1}{1 - 2x - 6} \\(g \circ f)(x) &= \frac{1}{-2x - 5}\end{aligned}$$

Ex.4 $f(x) = x^2 - 4$ and $g(x) = -2x + 1$ find $f(g(2x))$

$$\begin{aligned}f(g(2x)) &= f(-2(2x) + 1) \\f(g(2x)) &= f(-4x + 1) \\f(g(2x)) &= (-4x + 1)^2 - 4 \\f(g(2x)) &= (16x^2 - 8x + 1) - 4 \\f(g(2x)) &= 16x^2 - 8x - 3\end{aligned}$$

Ex.5 $f(x) = 3x - 1$, $g(x) = x^2$, $h(x) = \frac{1}{x}$ find $h \circ g \circ f$

$$\begin{aligned}\text{here } h \circ g \circ f &= h(g(f(x))) \\g(f(x)) &= (3x - 1)^2 \\h(g(f(x))) &= \frac{1}{(3x - 1)^2}\end{aligned}$$

Ex. 6 Find $f(g(h(x)))$ where $f(x) = x^2 + 5$, $g(x) = \frac{1}{x^2 - 2}$ $h(x) = \sqrt{x + 3}$

$$f(g(h(x))) = f(g(\sqrt{x + 3})) = f\left(\frac{1}{(\sqrt{x + 3})^2 - 2}\right)$$

$$f\left(\frac{1}{x + 3 - 2}\right) = f\left(\frac{1}{x + 1}\right) = \left(\frac{1}{x + 1}\right)^2 + 5$$

$$f(g(h(x))) = \frac{1^2}{(x + 1)^2} + 5 = \frac{1}{x^2 + 2x + 1} + 5$$

$$f(g(h(x))) = \frac{1}{(x + 1)(x + 1)} + 5$$

* Example of Composition of Functions

Q.1 Given the functions $f(x)=3x-5$ and $g(x)=x^2+2$, evaluate for:

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\(f \circ g)(2) &= f(g(2)) = 3(2^2 + 2) - 5 \\&= f(6) = 3(6) - 5 = 13\end{aligned}$$

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\&= [3x - 5]^2 + 2 \\(g \circ f)(-1) &= g(f(-1)) = [3(-1) - 5]^2 + 2 \\&= g(-8) = [-8]^2 + 2 \\&= 66\end{aligned}$$

Q.2 If $f(x) = 3x^2$, then find $(f \circ f)(x)$.

Solution: Given: $f(x) = 3x^2$

$$\begin{aligned}(f \circ f)(x) &= f(f(x)) \\&= f(3x^2) \\&= 3(3x^2)^2 \\&= 3(9x^4) \\&= 27x^4\end{aligned}$$

Q.3: If $f(x) = 2x$ and $g(x) = x+1$, then find $(f \circ g)(x)$ if $x = 1$.

Solution: Given, $f(x) = 2x$

$$g(x) = x + 1$$

Therefore, the composition of f from g will be;

$$(f \circ g)(x) = f(g(x)) = f(x+1) = 2(x+1)$$

Now putting the value of $x = 1$

$$f(g(1)) = 2(1+1) = 2(2) = 4$$

Q.4: If $f(x) = 2x + 1$ and $g(x) = -x^2$, then find $(g \circ f)(x)$ for $x = 2$.

Solution: Given,

$$f(x) = 2x + 1$$

$$g(x) = -x^2$$

To find: $g(f(x))$

$$g(f(x)) = g(2x+1) = -(2x+1)^2$$

Now put $x = 2$ to get;

$$g(f(2)) = -(2(2)+1)^2$$

$$\begin{aligned}
 &= -(4+1)^2 \\
 &= -(5)^2 \\
 &= -25
 \end{aligned}$$

Q.5: If there are three functions, such as $f(x) = x$, $g(x) = 2x$ and $h(x) = 3x$. Then find the composition of these functions such as $[f \circ (g \circ h)](x)$ for $x = -1$.

Solution: Given,

$$f(x) = x, \quad g(x) = 2x, \quad h(x) = 3x$$

To find: $[f \circ (g \circ h)](x)$

$$\begin{aligned}
 [f \circ (g \circ h)](x) &= f \circ (g(h(x))) \\
 &= f \circ g(3x) \\
 &= f(2(3x)) \\
 &= f(6x) \\
 &= 6x
 \end{aligned}$$

If $x = -1$, then;

$$[f \circ (g \circ h)](-1) = 6(-1) = -6$$

Q.6 $f(x) = \frac{1}{x+1}$, $g(x) = \frac{\sqrt{x-1}}{x}$, $h(x) = x^2 + 1$ than find $f \circ g \circ h$

Solution:

$$f \circ g \circ h = f[g(h)]$$

I will start by finding the composition $g \circ h = g(h)$

$$\begin{aligned}
 g(x) &= \frac{\sqrt{x-1}}{x} \\
 g[h(x)] &= \frac{\sqrt{h(x)}-1}{h(x)} \\
 &= \frac{\sqrt{(x^2+1)}-1}{x^2+1} \\
 &= \frac{\sqrt{x^2+1}-1}{x^2+1} \\
 &= \frac{\sqrt{x^2}}{x^2+1}
 \end{aligned}$$

$$g[h(x)] = \frac{x}{x^2+1} \longleftarrow \text{This intermediary function will be the input of the main function } f(x).$$

$$f(x) = \frac{1}{x+1}$$

$$f[g(h)] = \frac{1}{g(h)+1}$$

$$= \frac{1}{\frac{x}{x^2+1}+1}$$

$$= \frac{1}{\frac{x}{x^2+1}+1\left(\frac{x^2+1}{x^2+1}\right)}$$

$$= \frac{1}{\frac{x}{x^2+1}+\frac{x^2+1}{x^2+1}}$$

$$= \frac{1}{\frac{x^2+x+1}{x^2+1}}$$

$$= 1 \cdot \frac{x^2+1}{x^2+x+1}$$

$$f[g(h)] = \frac{x^2+1}{x^2+x+1}$$

$$f \circ g \circ h = f[g(h)] = \frac{x^2+1}{x^2+x+1} \longleftarrow \text{Final answer!}$$

Q.7 $f(x) = \frac{2}{x+3}$, $g(x) = \frac{-3x-2}{x}$ than find $f \circ g$

$$f(x) = \frac{2}{x+3}$$

Write down the main function

$$f[g(x)] = \frac{2}{g(x)+3}$$

Substitute the input function

$$= \frac{2}{\frac{-3x-2}{x}+3}$$

$$= \frac{2}{\frac{-3x-2}{x}+3\left(\frac{x}{x}\right)}$$

Add the complex denominator using LCD

$$= \frac{2}{\frac{-3x-2}{x}+\frac{3x}{x}}$$

$$= \frac{2}{\frac{-3x-2+3x}{x}}$$

Combine like terms on the numerator of the complex denominator

$$= \frac{2}{\frac{-2}{x}}$$

Simplify

$$= 2 \cdot \frac{x}{-2}$$

Divide by multiplying the numerator to the reciprocal of the denominator

$$f[g(x)] = -x$$

Simplify by canceling common factors

$$f \circ g = f[g(x)] = -x \longleftarrow \text{Final answer!}$$

Floor function: the greatest integer that is less than or equal to x is called floor function. Floor function can be represented: $\text{floor}(x) = \lfloor x \rfloor$

Ex. $\lfloor 2.3 \rfloor = 2$, $\lfloor 3 \rfloor = 3$, $\lfloor 0.5 \rfloor = 0$, $\lfloor -0.99 \rfloor = -1$

Ceiling function: the greatest integer that is greater than or equal to x is called ceiling function. ceiling function can be represented: $\text{ceiling}(x) = \lceil x \rceil$

Ex. $\lceil 2.3 \rceil = 3$, $\lceil 4 \rceil = 4$, $\lceil 0.5 \rceil = 1$, $\lceil -0.99 \rceil = 0$

Absolute value function: A function $f: \mathbb{R} \rightarrow \mathbb{R}$; $f(x) = |x|$ is known as the absolute value(or modulus) function, where $|x|$ is the absolute value of x .

Ex. $x = 10$ then $|x| = 10$, if $x = -10$ then $|x| = 10$

Exponential function: An exponential function is a Mathematical function in the form $f(x) = a^x$, where “ x ” is a variable and “ a ” is a constant which is called the base of the function and it should be greater than 0.

Ex. $f(x) = 2^x$, $g(x) = 3^x$

$$f(x) = 2^x$$

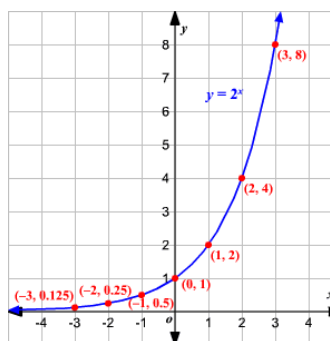
$$x = 0, f(0) = 1$$

$$x = 1, f(1) = 2$$

$$x = 2, f(2) = 4$$

$$x = -1, f(-1) = 0.5$$

$$x = -2, f(-2) = 0.25$$



Domain of $f(D_f) = (-\infty, \infty)$

Range of $f(R_f) = (0, \infty)$

Logarithm function: logarithm function is inverse function of exponential function.

i.e $f(x) = \log_a x$;

If $e^x = y$ then $x = \log_e y$ and $a^x = z$ then $x = \log_a z$

Ex. $2^3 = 8$ then $\log_2 8 = 3$

The logarithm properties are:

1. **Product Rule:**

The logarithm of a product is the sum of the logarithms of the factors.

$$\log_a xy = \log_a x + \log_a y$$

2. **Quotient Rule**

The logarithm of a quotient is the logarithm of the numerator minus the logarithm of the denominator.

$$\log_a \left(\frac{x}{y} \right) = \log_a x - \log_a y$$

3. **Power Rule**

$$\log_a x^n = n \log_a x$$

4. **Change of Base Rule**

$$\log_a b = \frac{\log_c b}{\log_c a}$$

$$\log_a b = \frac{1}{\log_b a}$$

where x and y are positive, and $a > 0$, $a \neq 1$

The following can be derived from the above properties.

$$\log_a 1 = 0$$

$$\log_a a = 1$$

$$\log_a a^r = r$$

$$\log_a \frac{1}{b} = -\log_a b$$

$$\log_{\frac{1}{a}} b = -\log_a b$$

$$\log_a b \log_b c = \log_a c$$

$$\log_{a^m} a^n = \frac{n}{m}, m \neq 0$$

Ex.1 Simplify $\log_6 24 + 2 \log_6 3$

Sol. $\log_6 24 + 2 \log_6 3$
 $= \log_6 24 + \log_6 3^2$
 $= \log_6 24 + \log_6 9$
 $= \log_6 (24 \times 9)$
 $= \log_6 216$
 $= 3$

Ex.3 $\log_7 (x^2 + 3x + 9) = 2$

Sol. $\log_7 (x^2 + 3x + 9) = 2$
 $7^2 = x^2 + 3x + 9$
 $49 = x^2 + 3x + 9$
 $40 = x^2 + 3x$
 $0 = x^2 + 3x - 40$
 $0 = (x + 8)(x - 5)$
 $x = -8, x = 5$

Ex. 5

$$\log_3 \left[\frac{18(x+2)^2}{(x-2)^3(x+5)^2} \right]$$

$$\begin{aligned} &= \log_3 [18(x+2)^2] - \log_3 [(x-2)^3(x+5)^2] \\ &= \log_3 (18) + \log_3 (x+2)^2 - [\log_3 (x-2)^3 + \log_3 (x+5)^2] \\ &= \log_3 (18) + \log_3 (x+2)^2 - \log_3 (x-2)^3 - \log_3 (x+5)^2 \\ &= \log_3 (9 \cdot 2) + 2 \log_3 (x+2) - 3 \log_3 (x-2) - 2 \log_3 (x+5) \\ &= \log_3 (9) + \log_3 (2) + 2 \log_3 (x+2) - 3 \log_3 (x-2) - 2 \log_3 (x+5) \\ &= 2 + \log_3 (2) + 2 \log_3 (x+2) - 3 \log_3 (x-2) - 2 \log_3 (x+5) \quad \checkmark \end{aligned}$$

© chilimath.com

Ex.2 $\log(x + 1) + \log(x - 1) = \log 8$

Sol. $\log(x + 1) + \log(x - 1) = \log 8$
 $\log(x^2 - 1) = \log 8$
 $x^2 - 1 = 8$
 $x^2 = 9$
 $x = 3$

Ex.4 $\log_4(2x + 6) - \log_4(x - 1) = 1$

Sol. $\log_4(2x + 6) - \log_4(x - 1) = 1$
 $\log_4 \left(\frac{2x+6}{x-1} \right) = 1$
 $\frac{2x+6}{x-1} = 4$
 $2x + 6 = 4x - 4$
 $x = 5$

Recursive Function: A function which calls it-self function is called recursive function.

Ex. $f(x) = f(x - 1) + 3$

$$h(t) = 3h(t - 1) + 11$$

$$v(x) = 2v(x - 1) + 3v(x - 2)$$